Height Growth Equations for CONUS: Douglas-fir

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Data

We took the FIA diameter growth data subset for Douglas-fir, filtered out stray (likely key-punch errors) states (NY, PA, IL, and MI), FIA plot locations with missing EPA L3 Codes, and where the diameter at the end of the 5-year or greater measurement interval was less than the starting diameter. This left 177860 growth observations.

The observations are distributed among L3 Codes as follows:

NA_L3NAME	NA_L3CODE	n
Cascades	6.2.7	41961
Coast Range	7.1.8	27597
Klamath Mountains	6.2.11	19329
Columbia Mountains/Northern Rockies	6.2.3	19118
Middle Rockies	6.2.10	14340
Blue Mountains	6.2.9	9893
North Cascades	6.2.5	9571
Southern Rockies	6.2.14	7481
Idaho Batholith	6.2.15	6632
Eastern Cascades Slopes and Foothills	6.2.8	4791
Strait of Georgia/Puget Lowland	7.1.7	3333
Sierra Nevada	6.2.12	3072
Canadian Rockies	6.2.4	2421
Wasatch and Uinta Mountains	6.2.13	1715
Arizona/New Mexico Mountains	13.1.1	1668
Willamette Valley	7.1.9	1541
Northwestern Great Plains	9.3.3	901
Colorado Plateaus	10.1.6	714
Northern Basin and Range	10.1.3	550

NA_L3NAME	NA_L3CODE	n
California Coastal Sage, Chaparral, and Oak Woodlands	11.1.1	504
Columbia Plateau	10.1.2	414
Central Basin and Range	10.1.5	99
Wyoming Basin	10.1.4	78
Madrean Archipelago	12.1.1	70
Northwestern Glaciated Plains	9.3.1	42
Southwestern Tablelands	9.4.3	20
Arizona/New Mexico Plateau	10.1.7	4
Snake River Plain	10.1.8	1

Diameter Growth Prediction

We built a data set for from Forest Inventory and Analysis (FIA) remeasurement data, filtering out observations with missing values for diameter at breast height (`dbh`), basal area in larger trees (`bal`), crown ratio (`cr`), total height (`ht`), site index (`si`), summer precipitation (PPT_{sm}), extreme minimum temperature (`EMT`), and remeasurement intervals less than 5 years. We also restricted the data to species with $\geq$ 5000 observations.

The diameter growth equation fit to these data was:

$$\Delta dbh = e^{(\beta_0 + \beta_1 log(\frac{(dbh+1)^2}{(cr*ht+1)^{\beta_4}}) + \beta_2 \frac{bal^{\beta_5}}{dbh+2.7} + \beta_3 PPT_{sm} + \beta_6 EMT)} \tag{1}$$

where:

- dbh = diameter at breast height (inches)
- cr = crown ratio (fraction of total height)
- ht = total height (feet)
- bal = basal area in larger trees ($feet^2/acre$)
- PPT_{sm} = summer precipitation (mm)
- EMT = extreme minimum temperature ($^\circ C$)

Equation 1 was fit to each qualifying species using an integrated fitting approach and errors minimized were ending `dbh`.

The parameter estimates and fit statistics are:

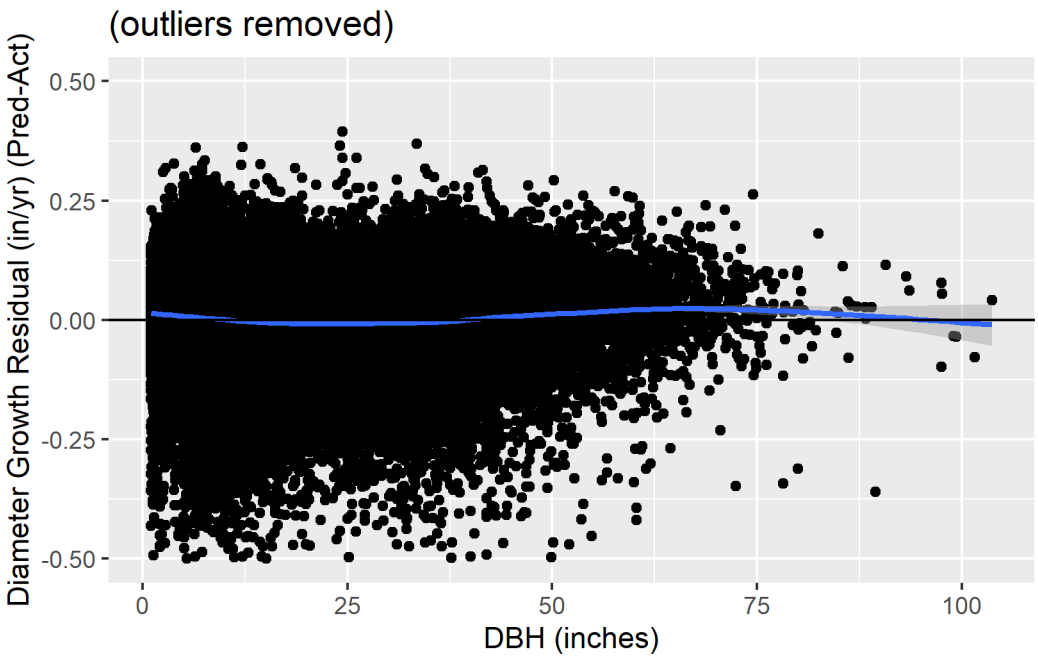
	Coef.	Std. error	t-stat.	p
b0	-1.3947486	0.0175903	-79.29070	0
b1	-0.5999958	0.0024630	-243.60617	0
b2	-0.1014764	0.0045171	-22.46497	0

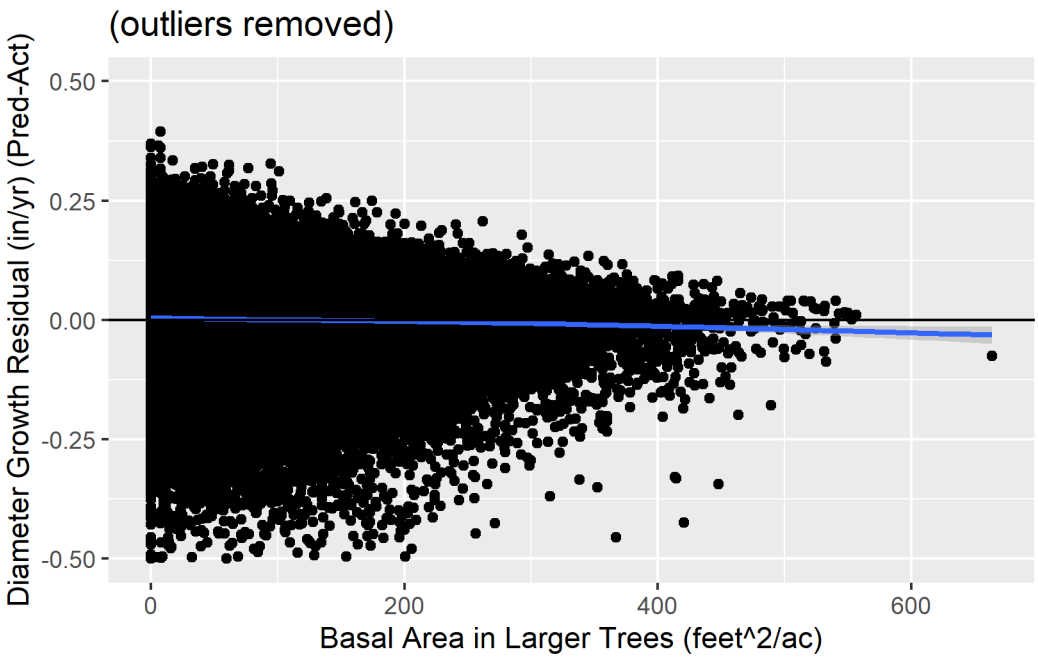
	Coef.	Std. error	t-stat.	p
b3	0.0007642	0.0000274	27.90307	0
b4	1.8694725	0.0060441	309.30600	0
b5	0.6063889	0.0078777	76.97582	0
b6	0.0424720	0.0001889	224.84009	0

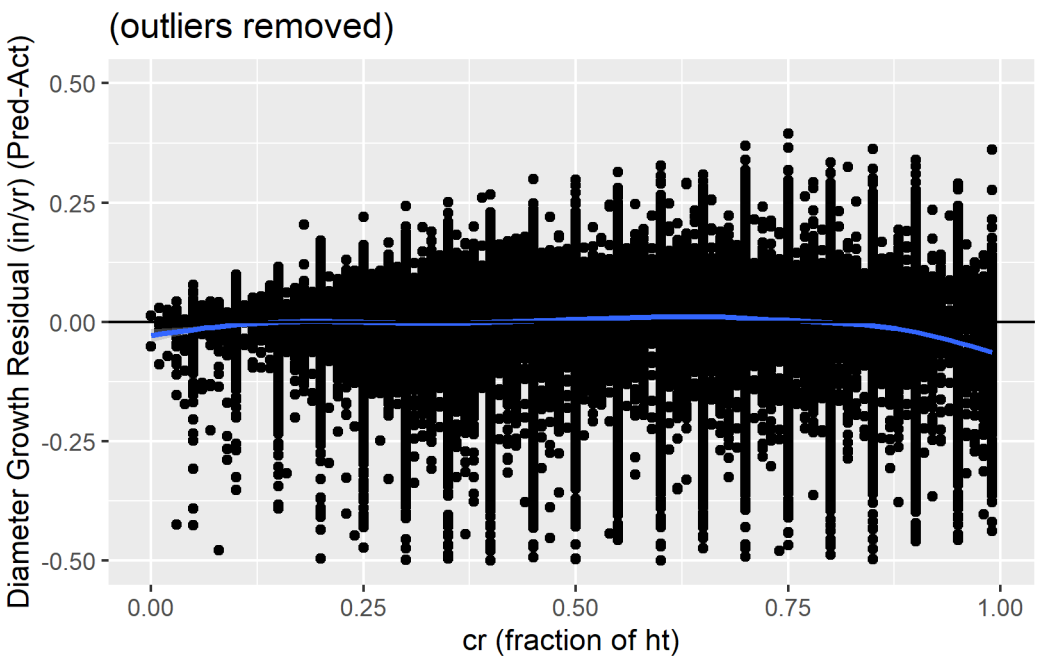
```
Residual Standard Error: 0.821999497400093 on 177853 degrees of freedom. AIC =  435027.2
```

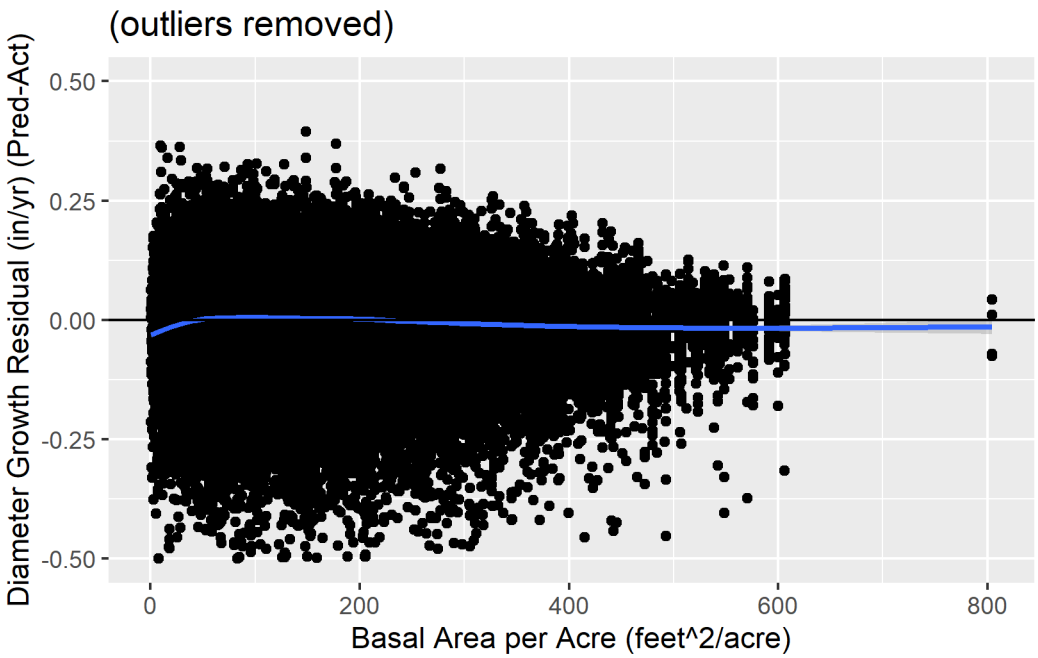
Residual Analysis for Equation 1

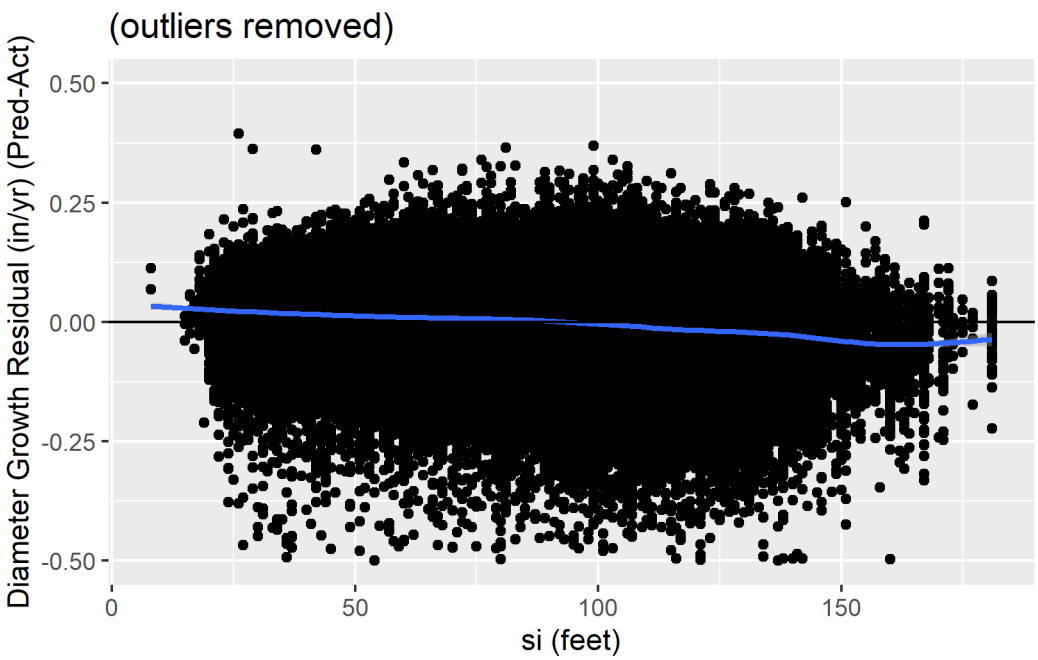
Mensurational Variables

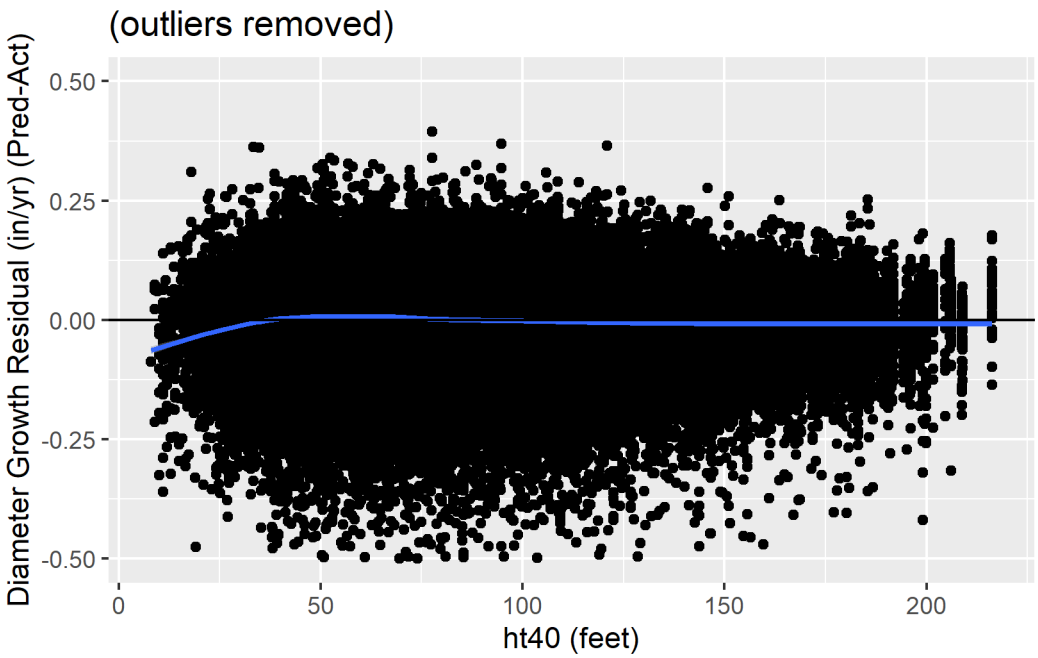



(outliers removed)
Diameter Growth Residual (in/yr) (Pred-Act)
0.50
0.25
0.00
-0.25
-0.50
0
200
400
600
Basal Area in Larger Trees (feet^2/ac)

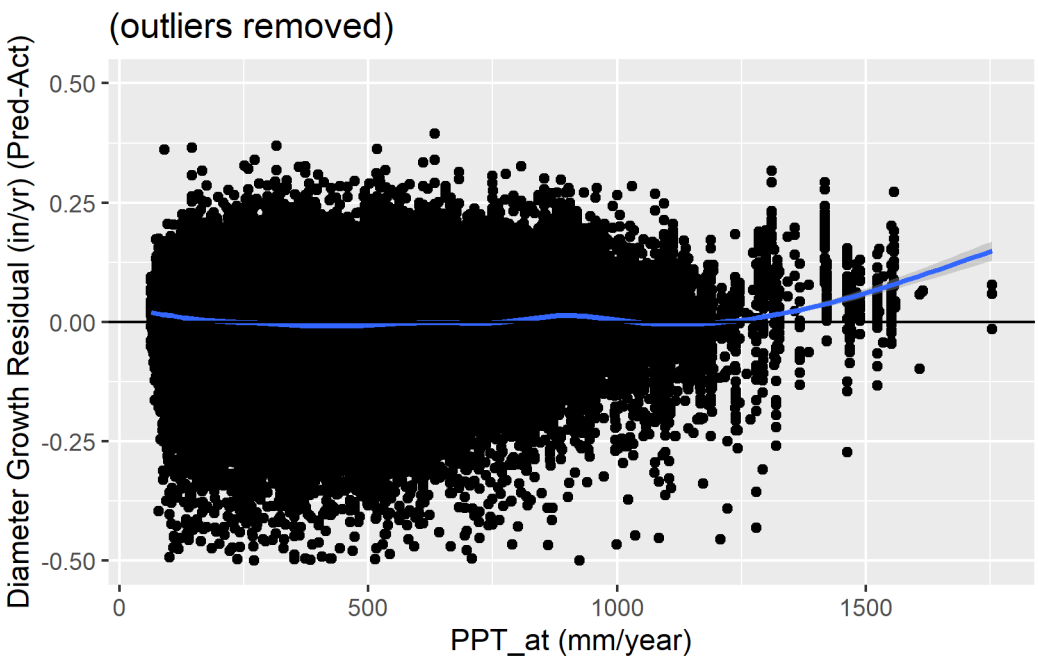

(outliers removed)
Diameter Growth Residual (in/yr) (Pred-Act)
0.50
0.25
0.00
-0.25
-0.50
0.00
0.25
0.50
0.75
1.00
cr (fraction of ht)

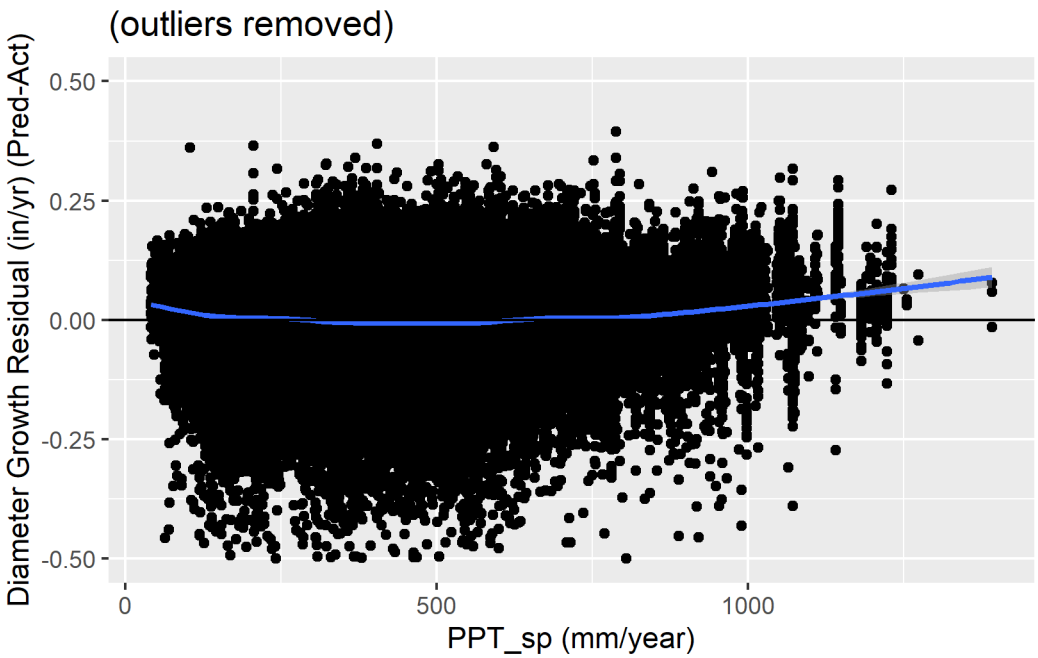

(outliers removed)
Diameter Growth Residual (in/yr) (Pred-Act)
0.50
0.25
0.00
-0.25
-0.50
0
200
400
600
800
Basal Area per Acre (feet^2/acre)

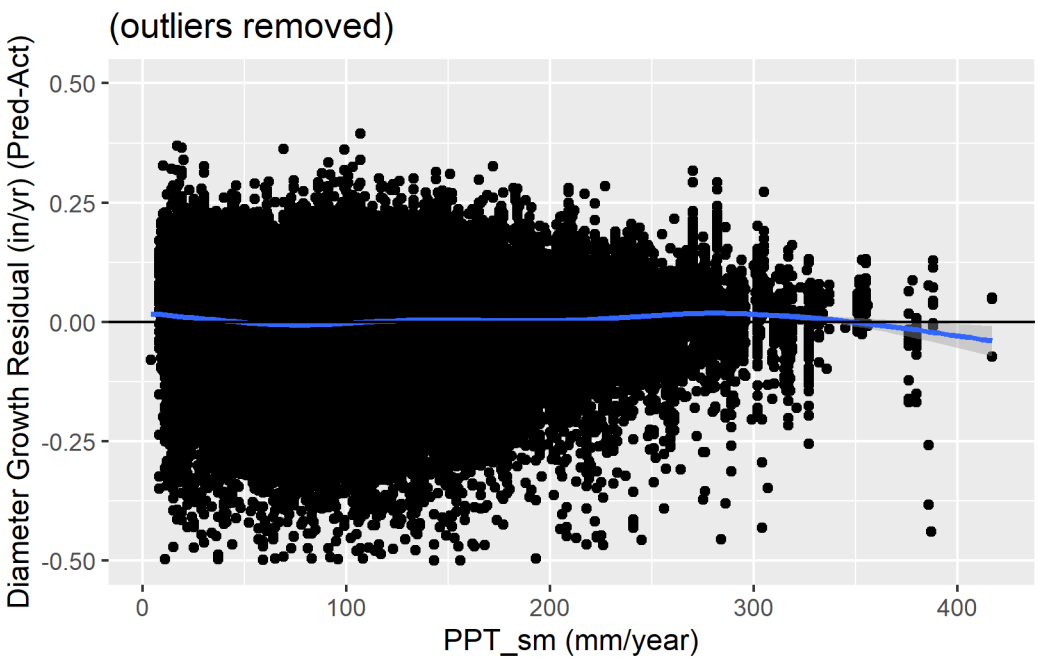

(outliers removed)
Diameter Growth Residual (in/yr) (Pred-Act)
0.50
0.25
0.00
-0.25
-0.50
0
50
100
150
si (feet)

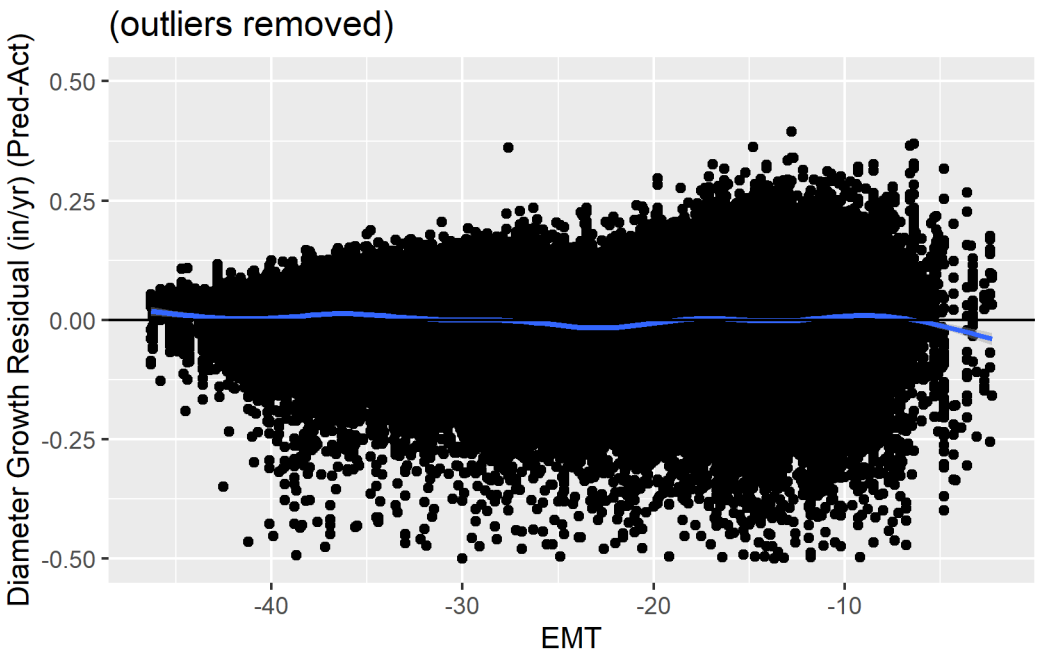


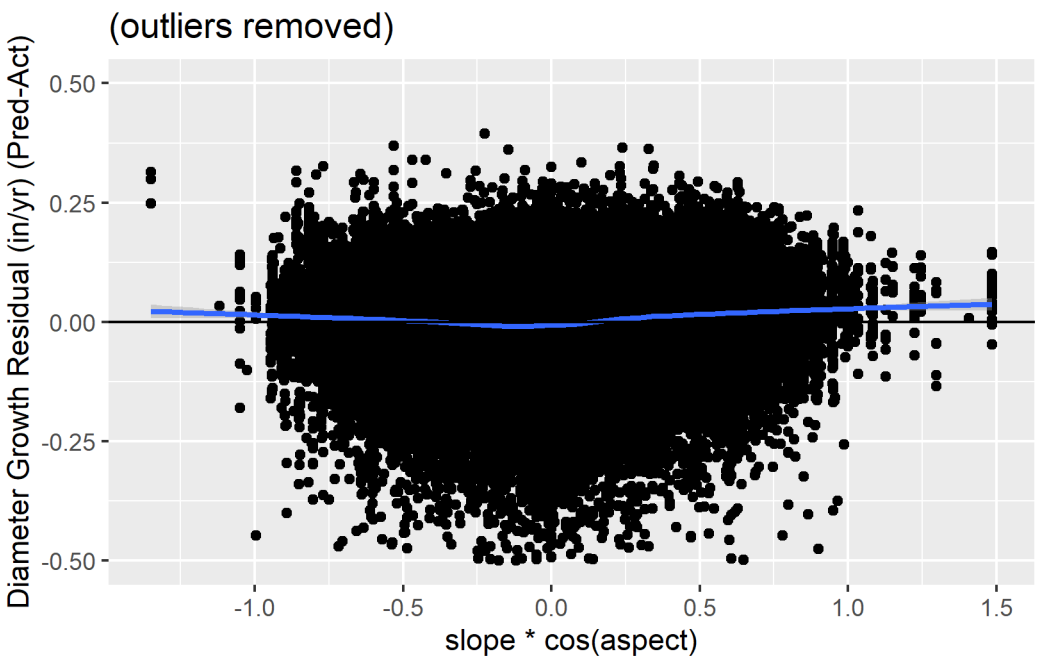
Environmental and Climate Variables

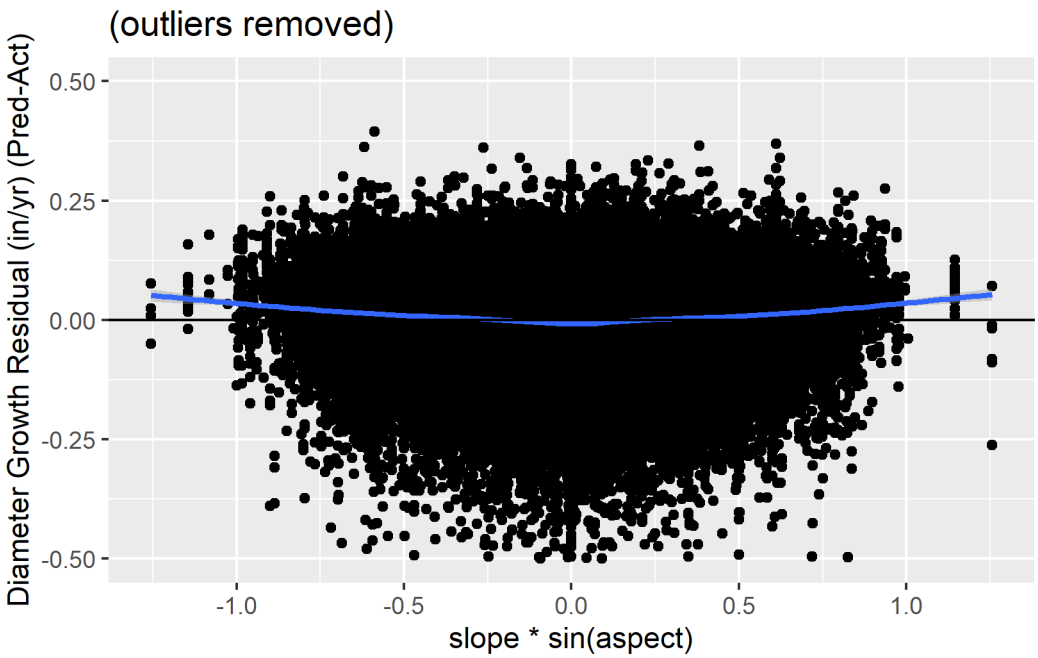


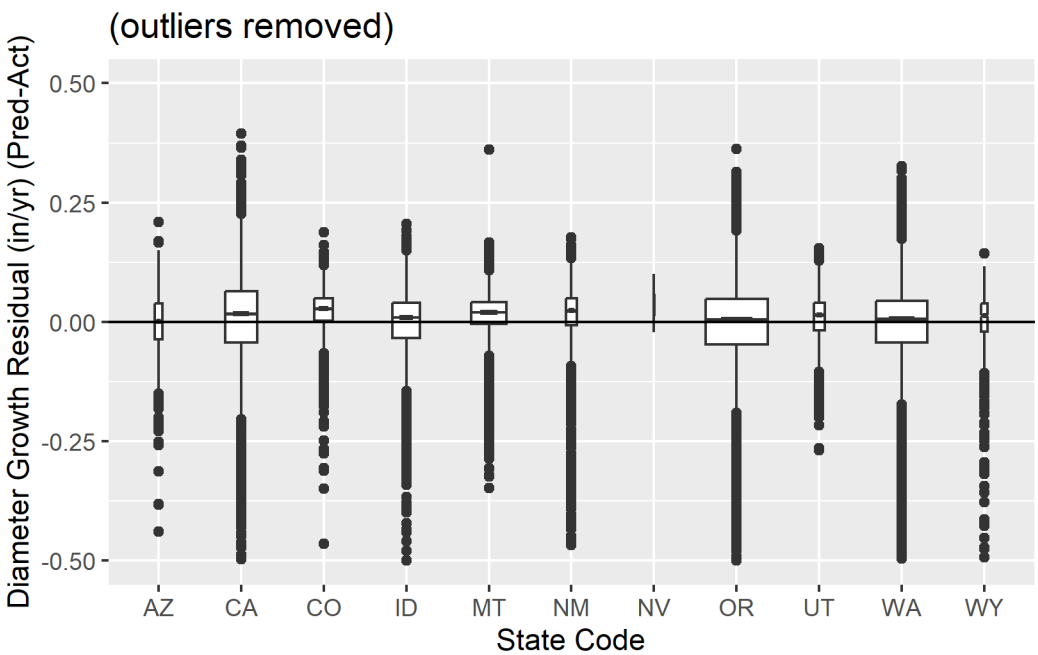

(outliers removed)
Diameter Growth Residual (in/yr) (Pred-Act)
0.50
0.25
0.00
-0.25
-0.50
0
500
1000
PPT_sp (mm/year)

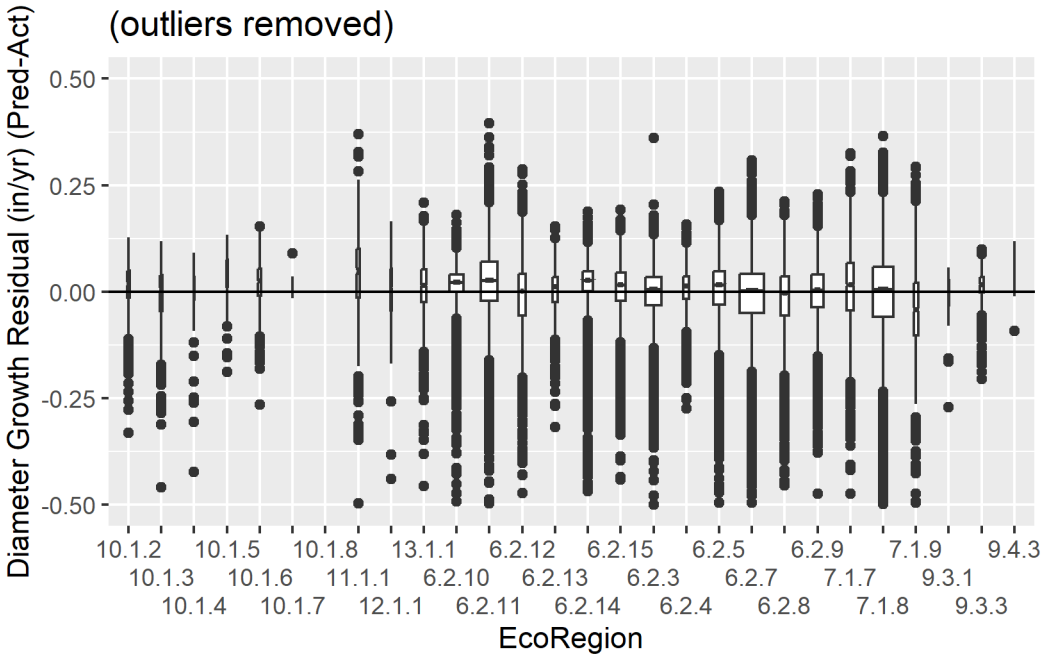

(outliers removed)
Diameter Growth Residual (in/yr) (Pred-Act)
0.50
0.25
0.00
-0.25
-0.50
0
100
200
300
400
PPT_sm (mm/year)

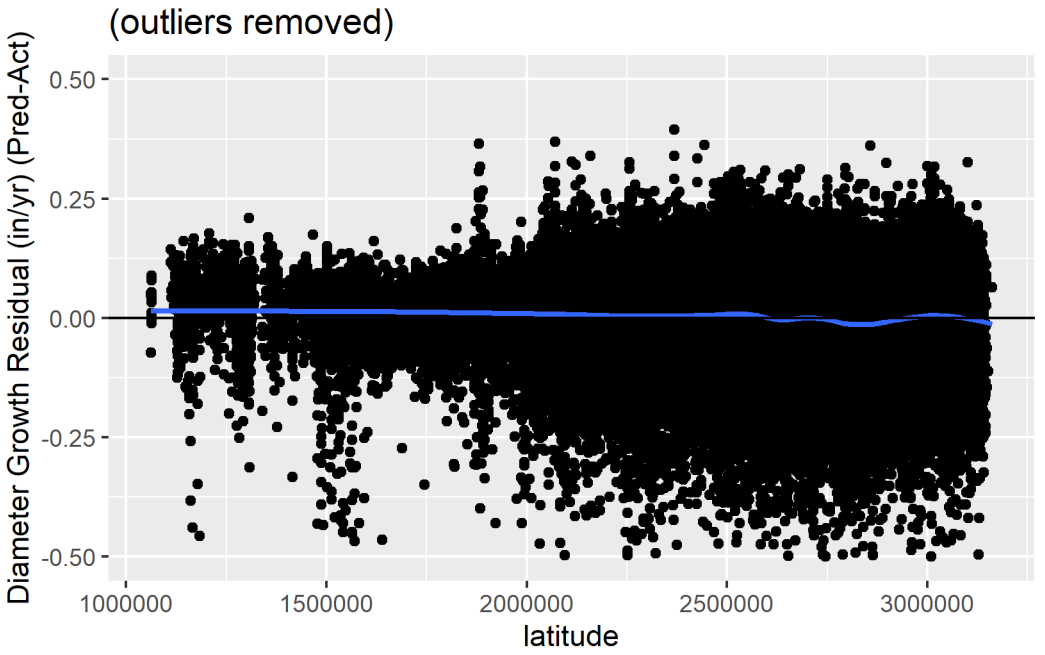

(outliers removed)
Diameter Growth Residual (in/yr) (Pred-Act)
0.50
0.25
0.00
-0.25
-0.50
-40
-30
-20
-10
EMT

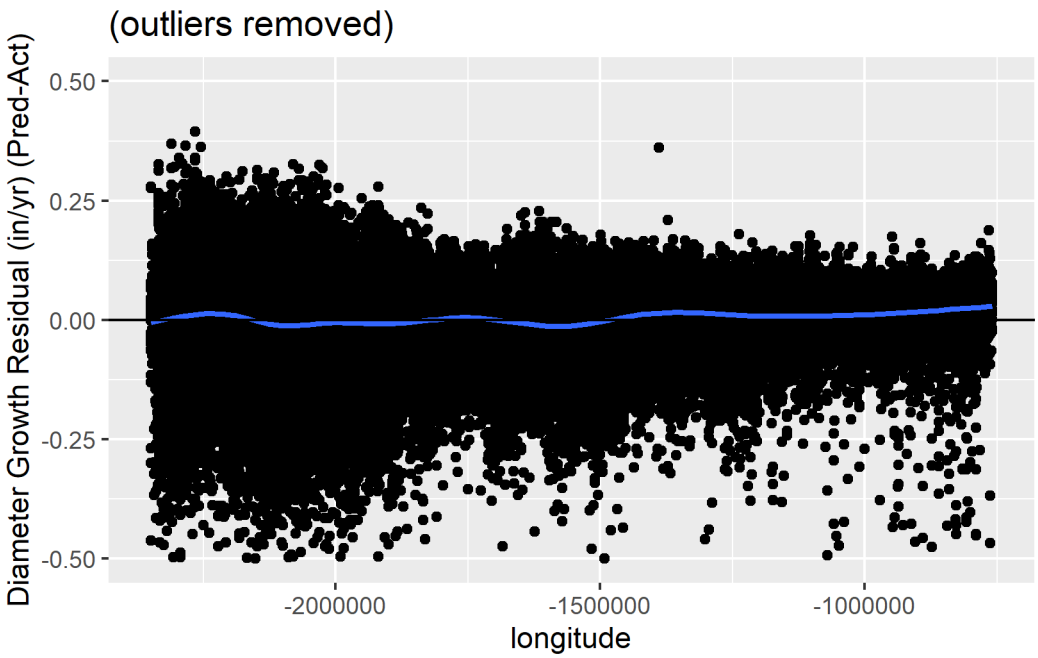

(outliers removed)
Diameter Growth Residual (in/yr) (Pred-Act)
0.50
0.25
0.00
-0.25
-0.50
-1.0
-0.5
0.0
0.5
1.0
1.5
slope * cos(aspect)

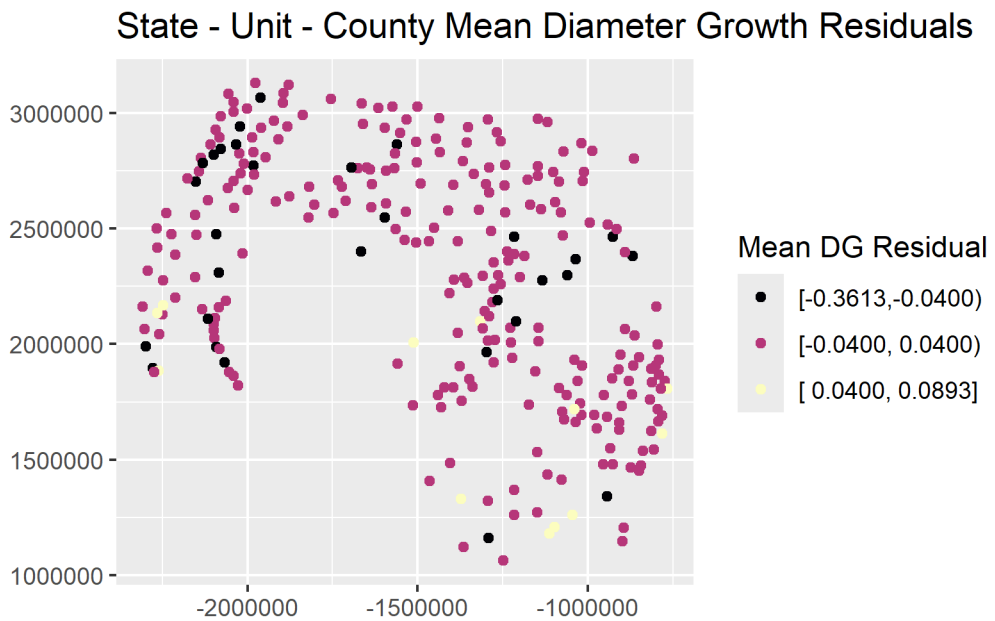

(outliers removed)
Diameter Growth Residual (in/yr) (Pred-Act)
0.50
0.25
0.00
-0.25
-0.50
-1.0
-0.5
0.0
0.5
1.0
slope * sin(aspect)


(outliers removed)
Diameter Growth Residual (in/yr) (Pred-Act)
0.50
0.25
0.00
-0.25
-0.50
AZ
CA
CO
ID
MT
NM
NV
OR
UT
WA
WY
State Code

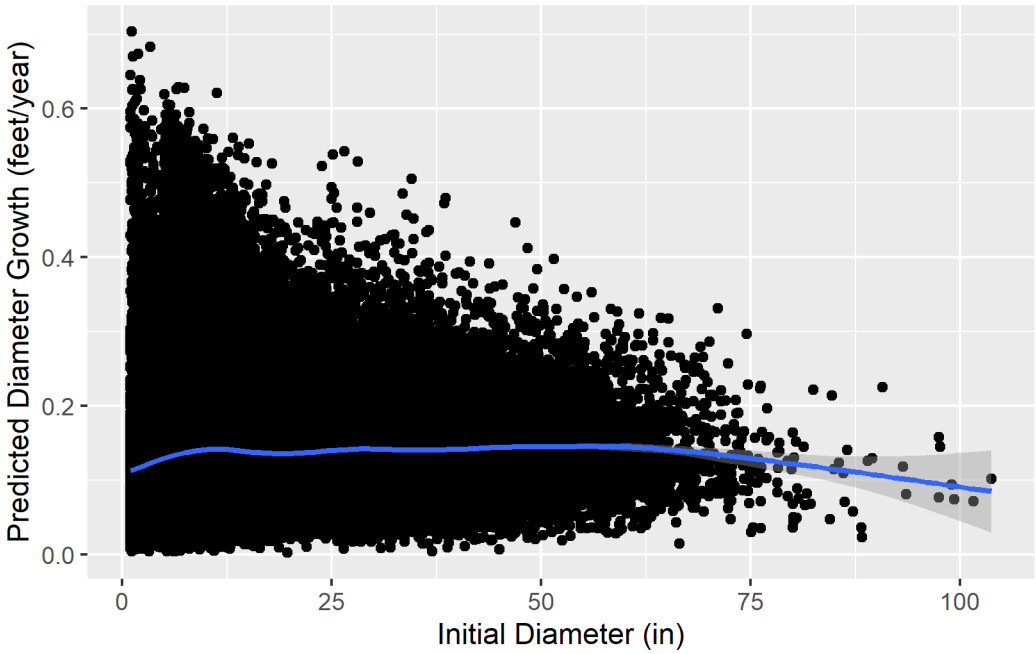

(outliers removed)
Diameter Growth Residual (in/yr) (Pred-Act)
0.50
0.25
0.00
-0.25
-0.50
10.1.2
10.1.3
10.1.4
10.1.5
10.1.6
10.1.7
10.1.8
11.1.1
12.1.1
13.1.1
6.2.10
6.2.11
6.2.12
6.2.13
6.2.14
6.2.15
6.2.3
6.2.4
6.2.5
6.2.7
6.2.8
6.2.9
7.1.7
7.1.8
7.1.9
9.3.1
9.3.3
9.4.3
EcoRegion

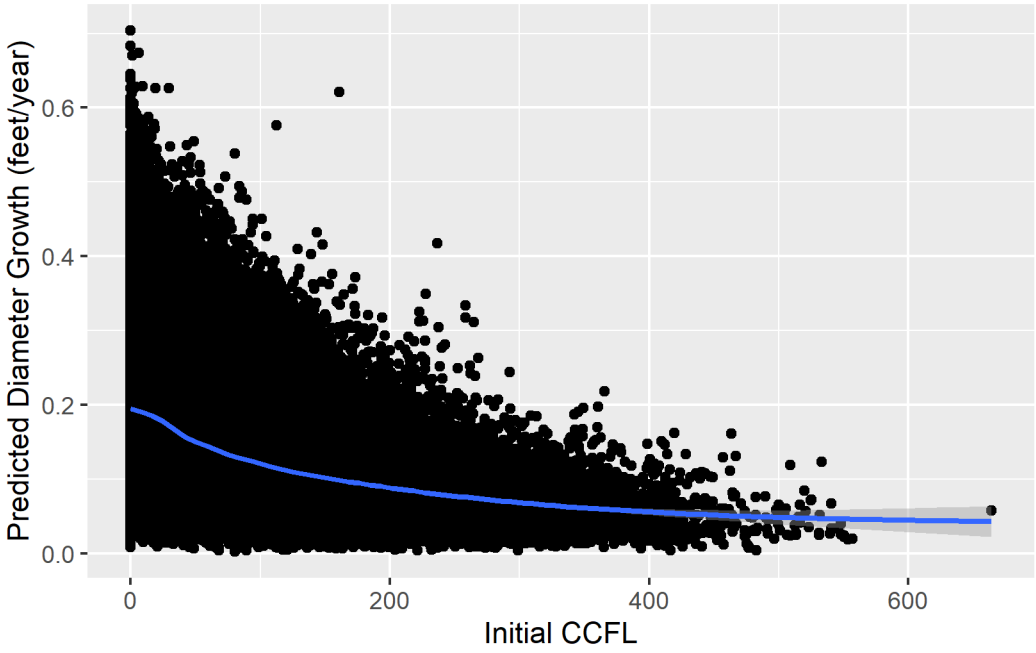

(outliers removed)
Diameter Growth Residual (in/yr) (Pred-Act)
0.50
0.25
0.00
-0.25
-0.50
1000000
1500000
2000000
2500000
3000000
latitude

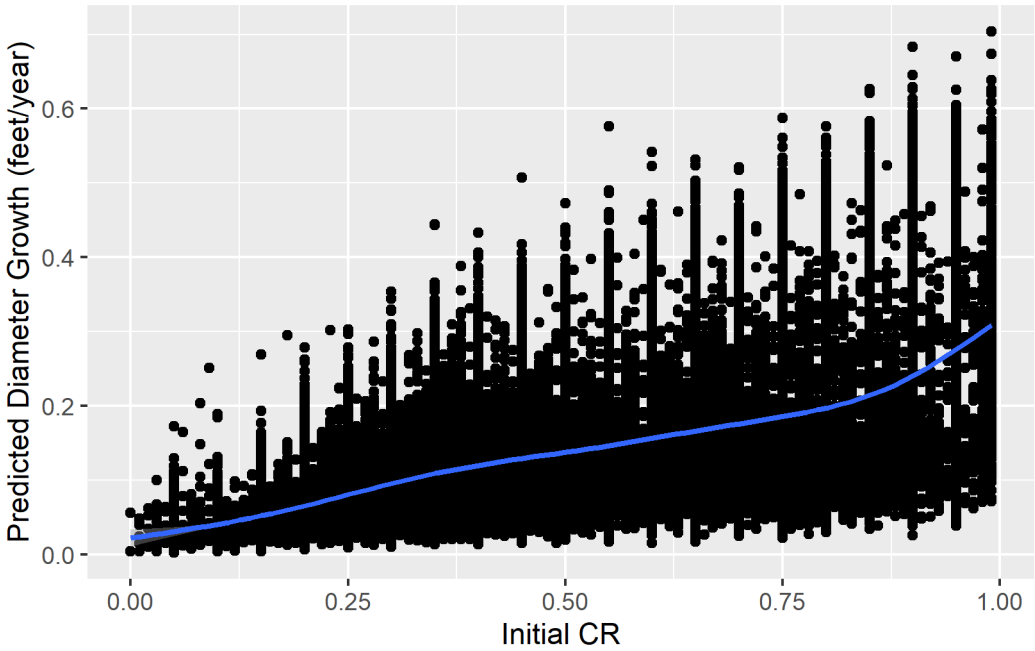

(outliers removed)
Diameter Growth Residual (in/yr) (Pred-Act)
0.50
0.25
0.00
-0.25
-0.50
-2000000
-1500000
-1000000
longitude

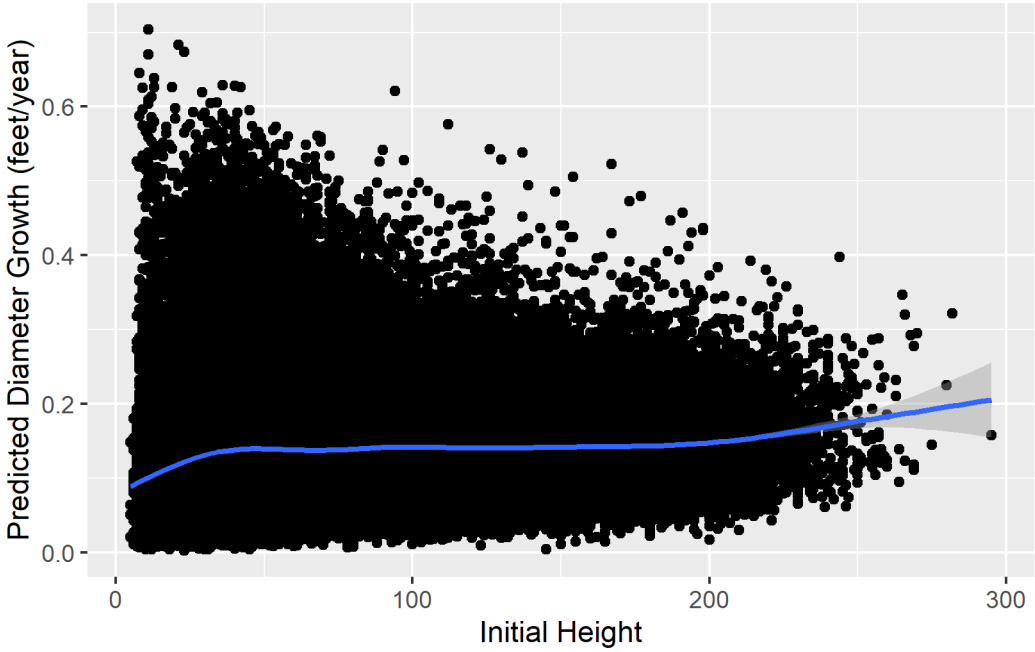

State - Unit - County Mean Diameter Growth Residuals
3000000
2500000
2000000
1500000
1000000
-2000000
-1500000
-1000000
Mean DG Residual
[-0.3613,-0.0400)
[-0.0400, 0.0400)
[0.0400, 0.0893]

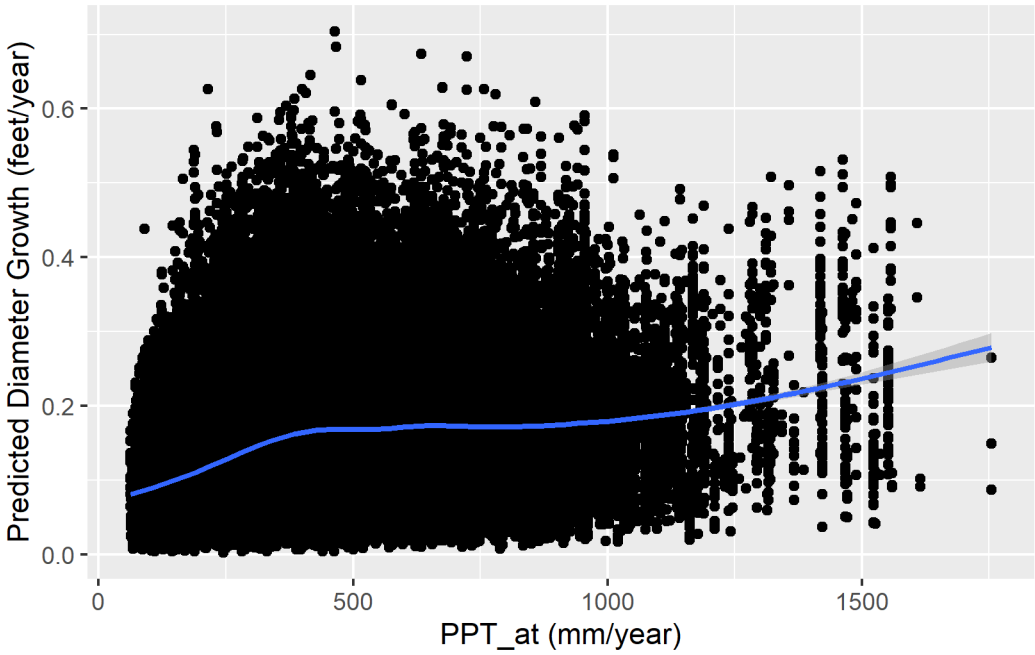
Prediction Performance

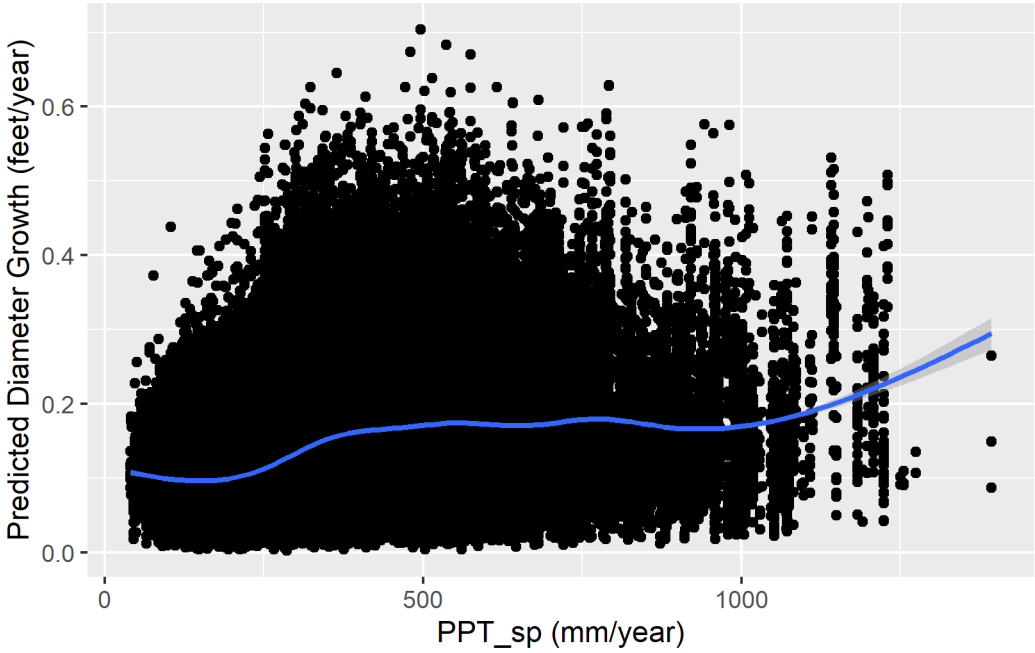


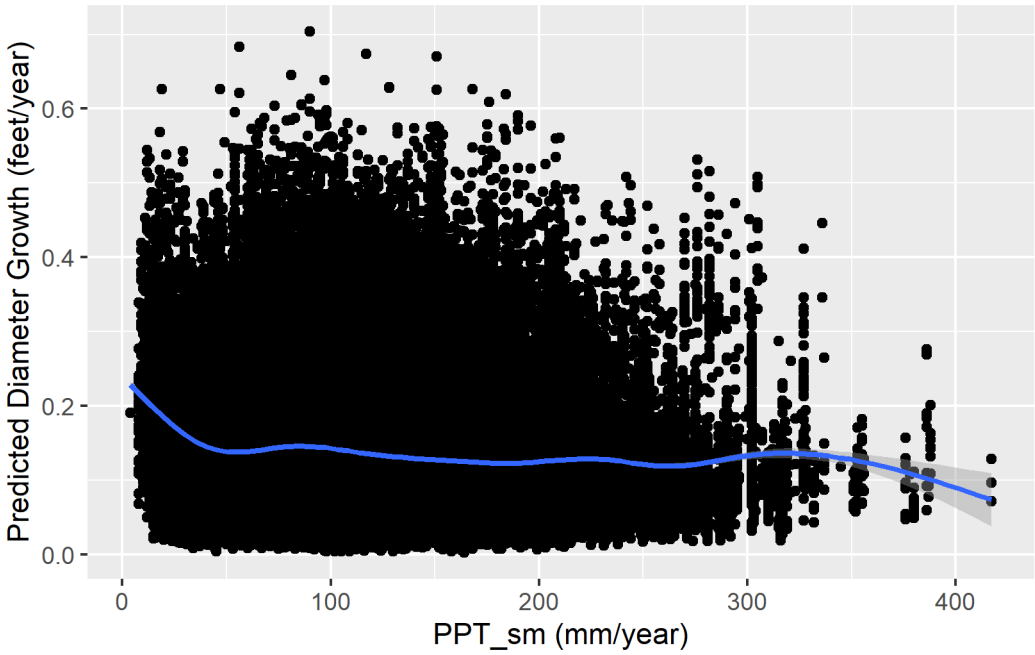


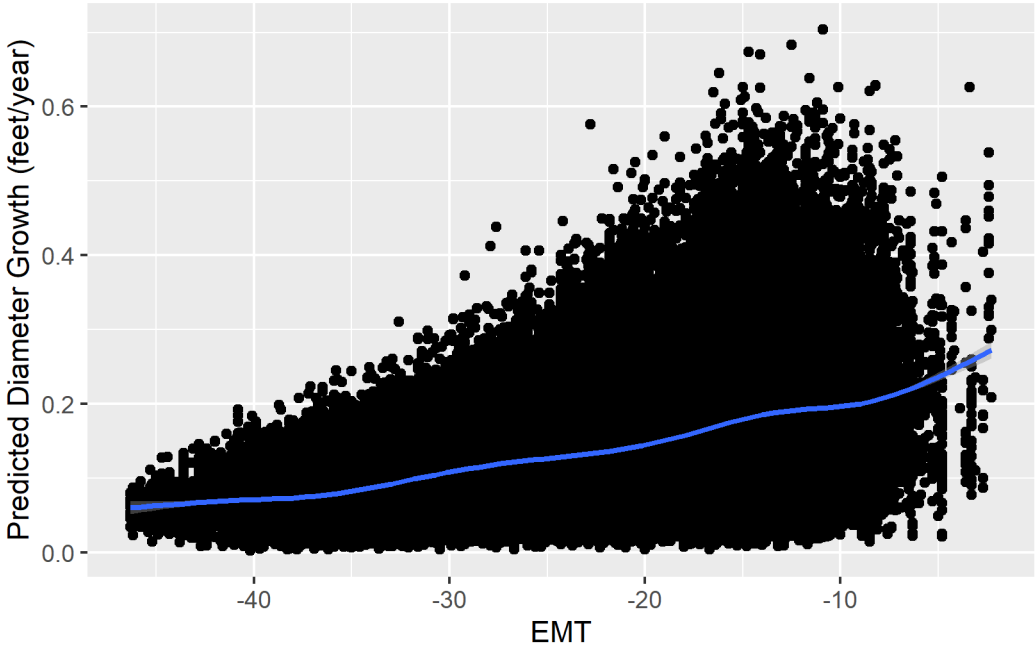

Predicted Diameter Growth (feet/year)
0.0
0.2
0.4
0.6
0.00
0.25
0.50
0.75
1.00
Initial CR


Predicted Diameter Growth (feet/year)
0.0
0.2
0.4
0.6
0
100
200
300
Initial Height


Predicted Diameter Growth (feet/year)
0.0
0.2
0.4
0.6
0
500
1000
1500
PPT_at (mm/year)


Predicted Diameter Growth (feet/year)
0.0
0.2
0.4
0.6
0
500
1000
PPT_sp (mm/year)


Predicted Diameter Growth (feet/year)
0.0
0.2
0.4
0.6
0
100
200
300
400
PPT_sm (mm/year)


Predicted Diameter Growth (feet/year)
0.0
0.2
0.4
0.6
-40
-30
-20
-10
EMT